Thesis Draft: HIV Care Cascade + Data Science Capacity Building

Strengthening the HIV Care and Prevention Cascade for Key Populations in Nigeria, and Building Data Science Capacity for Global Health

Thèse présentée à la Faculté de Médecine de l'Université de Genève pour obtenir le grade de Docteur en Sciences Biomédicales, Mention Santé Globale

par **Kene David Nwosu**

de **Nigeria**

Thèse n° [XXX]

Thèse préparée sous la direction du **Professeur Olivia Keiser**

Geneva, 2026

Acknowledgments

[To be completed]

List of Publications

Published articles

1. **Nwosu KD**, Kalaiwo A, Ng'ambi W, Oyelaran O, Umoh P, Estill J, Keiser O. Oral PrEP continuation rates among key populations in Nigeria: a retrospective cohort study of a large-scale HIV prevention programme. *BMJ Global Health* 2025; 10(12): e018739. doi: 10.1136/bmjgh-2024-018739

2. **Nwosu KD**, Kalaiwo A, Ngambi WF, Estill J, Omo-Emmanuel UK, Emmanuel G, Keiser O. Factors associated with viral load testing and viral suppression among HIV-positive female sex workers in Nigeria. *PLoS ONE* 2024; 19(5): e0304487. doi: 10.1371/journal.pone.0304487

Manuscripts in preparation

3. **Nwosu KD**, [co-authors to be finalized]. R Basics and Beyond: Outcomes of a Cohort-Based Online Programming Course for Health Professionals. *[Target journal: Cogent Education]*.

In this thesis, these works are referred to as **Paper 1** (PrEP continuation; *BMJ Global Health* 2025), **Paper 2** (viral load testing and suppression among female sex workers; *PLoS ONE* 2024), and **Paper 3** (R Basics and Beyond bootcamp evaluation; in preparation), and are also referred to throughout as **the PrEP analysis**, **the viral suppression analysis**, and **the R course evaluation**, respectively.

Abstract

Background

Key populations, including female sex workers (FSWs), men who have sex with men (MSM), people who inject drugs (PWID), and transgender individuals, bear a disproportionate burden of HIV infection. Improving outcomes for these populations requires understanding where the HIV care and prevention cascades are succeeding and where they are failing. Large-scale programmes serving key populations increasingly collect the routine data needed to answer these questions, but much of this data remains underutilized because health systems in the most affected countries lack sufficient analytical capacity [verify and cite]. Strengthening the HIV response therefore depends not only on generating programmatic evidence, but on building the data science workforce that can produce it.

Objectives

This thesis addresses both needs. Papers 1 and 2 analyse routine programmatic data from Nigeria's National HIV Key Populations programme, examining (a) oral PrEP continuation rates and associated factors among key populations, and (b) factors associated with viral load testing and suppression among HIV-positive female sex workers initiating antiretroviral therapy. Such analyses require data science skills that remain scarce in the countries most affected by HIV. The R course evaluation therefore examines the flagship course of The Graph Courses, an online training platform developed to strengthen these skills among health professionals in low- and middle-income countries.

Methods

All three studies employed retrospective analyses of routinely collected data. Papers 1 and 2 analyzed programmatic data from Nigeria's KP CARE 1 programme using mixed-effects logistic regression with random intercepts for service delivery facilities and multiple imputation by chained equations for missing data. The R course evaluation analyzed administrative enrollment rosters and post-course survey responses from nine cohorts of

an online R programming bootcamp, using logistic regression for quantitative outcomes and thematic coding for open-ended survey responses.

Results

Among 43,788 key population clients initiating oral PrEP between 2020 and 2023, the 6-month continuation rate was only 11.5%, revealing a critical gap on the prevention side of the cascade. Transgender individuals had the lowest continuation rate (3.5%), while female sex workers had the highest (13.7%). Among 34,976 FSWs initiating ART between 2016 and 2022, 97.1% received at least one viral load test and 94.5% achieved viral suppression, with the likelihood of both outcomes improving over time. Advanced clinical stage at baseline was associated with lower odds of suppression, underscoring the importance of early diagnosis. Across nine cohorts of the R Basics and Beyond bootcamp, over half of enrolled health professionals completed the course, with high satisfaction; both pre-recorded video lessons and live workshop interaction were highly valued, while time constraints were the primary barrier.

Conclusions

Taken together, the three studies illustrate a cycle in which data science skills enable the generation of programmatic evidence, and that evidence in turn identifies where the HIV response must be strengthened. The treatment cascade for FSWs in Nigeria is performing well but depends on early diagnosis; the prevention cascade requires urgent attention, particularly for transgender individuals and younger clients. Scaling access to the analytical skills demonstrated in these studies, as the training programme aims to do, is essential for sustaining data-driven improvements in HIV programmes and in global health more broadly.

Résumé en français

[À compléter: traduction française du résumé anglais, 1-2 pages]

General Introduction

I. HIV PREVENTION AND CARE AMONG KEY POPULATIONS

The Global Burden of HIV and the Disproportionate Impact on Sub-Saharan Africa

Over four decades since its initial discovery, the human immunodeficiency virus (HIV) remains one of the most pressing global health challenges. In 2024, an estimated 1.3 million people newly acquired HIV, and 630,000 died from causes related to acquired immunodeficiency syndrome (AIDS), which is caused by the virus (1). Although annual new

infections are now down by over 60% since their peak in 1995 (1), HIV infection is chronic and lifelong, meaning that every person who acquires the virus and survives adds to the prevalent pool over time. As such, the global number of people living with HIV has continued to rise, from an estimated 26.6 million in 2000 to 40.8 million in 2024 (1). Overall, HIV/AIDS remains the fourth-largest infectious disease contributor to global disability-adjusted life-year (DALY) losses, after lower respiratory infections, diarrhoeal diseases, and malaria (2,3).

Sub-Saharan Africa (SSA) suffers a disproportionate share of this burden. Of the 40.8 million people worldwide living with HIV in 2024, 26.3 million (65%) are in sub-Saharan Africa, although the region contains only about 15% of the world's population (4). During the peak of the epidemic in the late 1990s and early 2000s, AIDS became the leading cause of adult mortality in the region (5). By 2005, adult HIV prevalence exceeded 1% in 37 of the 48 sub-Saharan African countries, and the worst-affected countries saw multi-decade declines in life expectancy (4). In Eswatini, for example, adult prevalence reached 27% and life expectancy dropped from 61 to 32 years (4,6), while in Botswana, prevalence peaked at 25% and life expectancy dropped from 65 to roughly 47 years (4,6).

Since that time, significant progress has been made against the epidemic, largely driven by the scale-up of HIV antiretroviral therapy (ART) across the region. Modern ART regimens, typically taken as daily pill formulations of multiple drugs, suppress viral replication to undetectable levels, preventing progression to AIDS and rendering the individual effectively unable to transmit HIV (7). The number of people receiving ART in SSA grew from approximately 5 million in 2010 to about 22 million in 2024. Over that same period, new HIV infections in SSA dropped by 55%, from 1.46 million to 650,000, and AIDS-related deaths dropped by 60%, from 940,000 to 380,000 (4). Despite this progress, the SSA region continues to bear a disproportionate share of the global burden, accounting for 65% of all people living with HIV, 50% of new infections, and 60% of AIDS-related deaths (4).

Key Populations in the HIV Epidemic

Key populations (KPs) are groups of people who, due to social or behavioral factors, have a higher risk and prevalence of HIV. These include sex workers, men who have sex with men (MSM), people who inject drugs (PWID), and people in prisons and other closed settings (8). For example, Korenromp et al. found that the risk of acquiring HIV was 23 times higher for MSM, 14 times higher for PWID, and 8.9 times higher for sex workers, compared with the general adult population (9). According to their estimates, KPs and their sex partners accounted for 55% of new adult HIV infections in 2022. This is an increase from 2010, when the figure was 44% (9). Thus, as general-population incidence declines, the global epidemic is increasingly concentrated among these KP groups.

The reasons for the concentration of HIV among KPs are multifaceted. Part of the elevated risk reflects greater exposure to higher-transmission-risk events. Predominant sexual practices among MSM and transgender women have higher per-act transmission probabilities, sex workers often have more frequent sexual exposures, including

unprotected sex, than the general population, and PWID face direct exposure through shared injecting equipment (10,11). But these risk factors are amplified by structural and societal conditions that reduce access to prevention and care services.

Chief among these is criminalization. As of 2025, 64 countries criminalized same-sex relations, 168 countries partly or wholly criminalized sex work, and 14 countries had laws that effectively criminalized transgender people (1). Such legal environments push KPs underground, exposing them to discrimination and violence, and deterring them from seeking testing and treatment (12–14). When diagnosis and treatment are delayed, individuals carry higher viral loads for longer, increasing the risk of onward transmission and sustaining the high circulation of HIV within these populations. Empirical studies confirm the link between criminalization and poorer HIV outcomes. For example, in a cross-national analysis of 194 countries, those that criminalized same-sex relations had 11 percentage points lower HIV status awareness and 8 percentage points lower viral suppression among people living with HIV, with similar deficits where sex work and drug use were criminalized (15). In one notable case, Vasireddy et al. reported that weekly visits to drop-in centres serving KPs fell dramatically during the 2023 Anti-Homosexuality Act debate in Uganda, with at least four centres closing due to safety incidents (16).

Because KPs have a high burden of HIV, play a disproportionate role in transmission, and face unique structural and societal barriers to healthcare access, effective global management of the HIV epidemic requires prevention and treatment programmes tailored to this population. A growing body of evidence supports moving care out of traditional hospital settings and into community-based clinics that offer bundled services, including pre-exposure prophylaxis (PrEP) for preventing infection, HIV testing services, concurrent STI screening and treatment, and antiretroviral therapy for those diagnosed with HIV (17,18). There is also support for combining these clinical services with legal and social support structures, so that barriers beyond the biomedical, such as violence, discrimination, and legal vulnerability, can be addressed in the same visit (17). Combining all such services under one roof, in so-called “One-Stop Shops,” is increasingly considered best practice for KP service delivery. Finally, peer-led and mobile outreach models extend these services beyond fixed clinics, actively seeking out KP communities at venues and hotspots to facilitate engagement with prevention and care (19). However, while these frameworks are widely endorsed, evidence on how they perform at scale remains limited, particularly from large programmatic settings in sub-Saharan Africa, where the HIV burden is greatest.

Key Populations in Nigeria and the Programmatic Response

Nigeria, the most populous country in Africa with over 220 million inhabitants, has the fourth-largest HIV epidemic globally (20,21). While the overall adult HIV prevalence is relatively low (estimated at 1.3% in the 2018 national household survey (22)), the epidemic is severely concentrated among KPs. The 2020 Integrated Biological and Behavioural Surveillance Survey (IBBSS), which surveyed 17,975 KP members across 12 states, found

HIV prevalence of 15.5% among FSWs, 25.0% among MSM, 28.8% among transgender individuals, and 10.9% among PWID, all far exceeding the general population rate (23).

A hostile legal environment for KPs in Nigeria compounds their burden of HIV disease. The Same-Sex Marriage Prohibition Act of 2014 criminalized same-sex relationships with penalties of up to 14 years' imprisonment (12,24), and its broad provisions also frequently affect transgender individuals, for whom no legal pathway to gender recognition exists (24). Drug use is very strongly penalized, with sentences of up to 25 years' imprisonment for certain substances under the National Drug Law Enforcement Agency Act, further restricting access to harm reduction services for PWID (25). Sex work occupies a grey legal zone: selling sex is not itself a criminal offence, but soliciting clients and procurement of sex are criminalized under Sections 223–225A of the Criminal Code Act (24,26). Together, these laws push KPs underground, raising barriers to HIV prevention, testing, and treatment (12,13,15).

In response to the concentrated HIV burden among key populations, the Key Population Community HIV Services Action and Response (KP-CARE 1) programme was established in Nigeria with support from USAID and implemented by Heartland Alliance LTD/GTE (27,28). The programme delivers HIV prevention, testing, care, and treatment services for FSWs, MSM, PWID, and transgender individuals across multiple Nigerian states through a differentiated service delivery model centred on community-based One-Stop Shops (OSS) and mobile outreach (27–29). These OSS provide HIV testing and counselling, ART initiation and refills, pre-exposure prophylaxis (PrEP), sexually transmitted infection screening and treatment, and psychosocial support in KP-friendly settings designed to reduce stigma and improve access. In addition, venue-based and hotspot outreach helps link clients from brothels, bars, and other gathering places into prevention and treatment services (18,27,29,30).

The KP-CARE 1 programme has an electronic data management system and database containing information on clients enrolled in prevention, testing and care services. This system is a rich source of large-scale routine programmatic data that can be used for both programme evaluations and research. Papers 1 and 2 in this thesis use this data to analyse outcomes along the HIV prevention and care cascades.

HIV Prevention: Pre-Exposure Prophylaxis and the Challenge of Continuation

Pre-exposure prophylaxis (PrEP) drugs are antiretroviral medications (ARVs) taken by HIV-negative people to prevent HIV acquisition. The most common PrEP medications consist of daily oral tablets of tenofovir disoproxil fumarate combined with either emtricitabine (FTC) or lamivudine (3TC) (31). Clinical trial and cohort evidence shows that tenofovir-based PrEP can substantially reduce HIV acquisition when taken consistently (32–34).

Despite their proven biological efficacy, the overall impact of PrEP regimens is limited by low uptake among key populations and low continuation after starting. In a global meta-analysis of 59 studies covering over 43,000 participants, 41% had discontinued PrEP within six months of starting, and among those continuing, 37.7% had suboptimal adherence

patterns. In sub-Saharan Africa, discontinuation was 47.5% (35). Commonly reported reasons for discontinuation included side effects (25 studies), low perceived risk of infection (21 studies), cost or lack of health insurance (10 studies), and inaccessible care (nine studies). Issues of cost and access are likely to be worse for key populations, who often face criminalization, stigma, and other barriers to routine health services.

In Nigeria, PrEP rollout for key populations is an important component of the national HIV prevention strategy (36). The KP-CARE-1 programme works to provide PrEP to eligible KP clients using a provider-facilitated approach in which individuals testing HIV-negative are offered oral PrEP as part of a comprehensive prevention package. The PrEP analysis examines data from over 43,000 KP clients who initiated PrEP through the KP-CARE-1 programme. The large sample size makes it one of the largest PrEP continuation studies among key populations in sub-Saharan Africa. We report the 6-month continuation rates and identify factors associated with sustained PrEP use across different KP groups.

The HIV Care Cascade: Viral Load Testing and Suppression

The HIV Care Cascade is a conceptual model used to monitor the progression of people with HIV through key stages of medical care. It includes testing and diagnosis, linkage to care, initiation of ART, adherence to treatment, viral load testing and viral suppression (37). Attrition happens at each stage of this cascade; understanding each point of attrition is important for evaluating and improving HIV programmes.

Viral suppression is defined as having a very low viral load, commonly < 1000 copies/mL, and is the main goal of ART. Successful viral suppression improves health outcomes and greatly reduces the risk of onward HIV transmission. It is reflected in the UNAIDS 95-95-95 targets, which are global goals targeting a minimum of 95% awareness of status among people living with HIV, 95% ART access among those diagnosed, and 95% viral suppression among those on ART. In 2024, global estimates were still below these targets, with an estimated 87% aware of their status, 77% of those accessing ART, and 73% of those on ART being virally suppressed (1).

Viral load testing and suppression are especially critical for female sex workers living with HIV because they have a high transmission potential, given their high number of sexual partners. Existing FSW cascade evidence in sub-Saharan Africa is limited and usually is not based on large programme datasets (38).

This thesis (paper 2) addresses this gap by analyzing routine programmatic data from nearly 35,000 FSWs enrolled in the Nigerian KP programme to identify factors associated with receipt of viral load testing and achievement of viral suppression.

II. DATA SCIENCE CAPACITY BUILDING FOR GLOBAL HEALTH

Bridging Public Health Research and Data Science Capacity

Data analysis skills are increasingly important for the global health workforce. The digitization of health systems, from electronic health records to surveillance and

monitoring platforms, generates large volumes of data that can be analysed to improve programme design and delivery, such as in the HIV studies presented in this thesis. Analysing these data sources requires practical data science skills, from cleaning routine datasets and running reproducible analyses to interpreting results for programme use.

These data science skills are still limited in many low- and middle-income countries (LMICs), where they are often needed most. As a result, large amounts of routine health data are under-analysed, and programmes in these countries often rely on external technical support for advanced analyses (39). As a native of Nigeria who has seen these gaps firsthand, helping reduce the shortage of health data professionals in Africa was a core motivation for my doctoral pursuits.

Thus, a large fraction of my time during my doctoral work has been engaged in building an educational platform and courses, under the banner of the GRAPH Courses, that are aimed at providing affordable and high-quality educational experiences to help build out data capacity in health progressionals globally.

Approaches to Data Science Education

Innovative approaches are required to effectively fill the data science skill gap in LMICs. Short in-person professional workshops are one popular format, wherein instructors deliver intensive instruction with hands-on practice, often over a few days to groups of health professionals (40). However, such workshops can be difficult to scale due to their time and financial costs and geographical limitations (41).

Massive open online courses are another useful education mode for filling the data skills gap. They provide world-class educational content at low or no cost to anyone with access to the internet (42). MOOC platforms like Coursera and EdX became very popular in the 2010s, but enthusiasm for this modality has reduced due to their notoriously low completion rates, often in the single digits (43,44). Without a live instructor to provide accountability when student motivation wavers, or support when students get stuck, most learners will quickly disengage and drop out (45,46).

These challenges are particularly relevant for programming and data science training. R, the programming language used for all three studies in this thesis, is powerful and free but has a steep learning curve, particularly for users without prior programming experience. For building programming fluency in such a language, self-paced MOOCs are unlikely to be successful, as they leave learners stranded when they encounter errors or lose motivation, but intensive in-person workshops are also not ideal, because they do not provide sufficient time for sustained practice.

Cohort-based online courses are a promising middle ground that blend the flexibility of self-paced courses with the accountability of in-person workshops. These programmes typically use a flipped classroom model: learners engage with asynchronous content (pre-recorded videos, readings, exercises) on their own schedule, then participate in scheduled live sessions for instructor-led problem-solving and peer interaction (47). The

asynchronous component provides flexibility and reduces total contact hours, while the synchronous component provides accountability, community, and direct support, all of which are often absent from purely self-paced courses (48). The rise in remote working during the COVID-19 pandemic created new opportunities to deliver such programmes at scale, because many learners in LMICs became familiar with the videoconferencing software needed for such sessions (49).

The GRAPH Network and The Graph Courses

The GRAPH Network (Global Research and Analyses for Public Health) is an interdisciplinary network of researchers and health professionals headquartered at the University of Geneva's Institute of Global Health. It was established in 2020 in collaboration with the WHO African Regional Office to support the sub-Saharan African COVID-19 response (50). Through the organization's work with partner countries, we observed firsthand the widespread use of manual processes for data processing, as few staff had been trained in statistical programming languages, limiting the speed and quality of analyses. Out of this recognition, I proposed and developed The GRAPH Courses (thegraphcourses.org) as a dedicated training platform for data science and AI in health and life sciences.

The platform delivers training through two complementary models. First, live cohort-based bootcamps use the flipped classroom approach described above: each week, learners complete several hours of asynchronous prework (pre-recorded video lessons, interactive coding exercises, and online quizzes), then attend a 2–3 hour live workshop featuring instructor-led demonstrations, collaborative small-group problem-solving, and peer interaction. The flagship bootcamp is R Basics and Beyond (RBB), which introduces R programming and data analysis for health research and is the subject of the R course evaluation in this thesis; additional bootcamps cover French-language R, intermediate R, beginner Python, and applied generative AI. Second, free self-paced versions of these courses—video lessons, exercises, and quizzes without live workshops or instructor interaction—allow learners anywhere in the world to study at their own pace at no cost. Two open-access textbooks, *Introduction to Data Analysis with R* (51) and *Introduction to Data Science with Python* (52), accompany the introductory R and Python courses respectively.

The R course evaluation (in manuscript) examines the educational outcomes of the R Basics and Beyond bootcamp across nine cohorts from 2023 to 2025. Using enrollment rosters and post-course survey data, our study assesses completion rates, participant satisfaction, and learner experience.

Thesis Objectives

This thesis has three specific objectives:

1. **To investigate oral PrEP continuation rates and associated factors among key populations in Nigeria** (Paper 1). Using data from over 43,000 KP clients who

initiated oral PrEP through the KP CARE 1 programme, this study examines 6-month continuation rates across different KP groups and identifies factors associated with sustained PrEP use.

2. **To examine factors associated with viral load testing and viral suppression among HIV-positive female sex workers initiating antiretroviral therapy in Nigeria** (Paper 2). Using routine programmatic data from nearly 35,000 FSWs enrolled in the National HIV KP programme, this study identifies demographic, clinical, and structural factors associated with receipt of viral load testing and achievement of viral suppression.

3. **To evaluate the educational outcomes of a cohort-based online R programming course for health professionals** (Paper 3). Using enrollment rosters and post-course survey data from nine cohorts of the R Basics and Beyond bootcamp, this study assesses completion rates, participant satisfaction, highlights, and challenges.

Methodological Contributions

Overview

I played a central role in all three studies comprising this thesis: leading the data analysis and manuscript preparation for the HIV-related studies (Papers 1 and 2), and founding and leading the development of the training platform and evaluation study (Paper 3). This section details my specific contributions to each study.

Paper 1: PrEP Continuation Among Key Populations

For Paper 1, I led the data analysis and manuscript writing. This involved analysing data from over 43,000 PrEP clients, implementing mixed-effects logistic regression with random intercepts for service delivery sites. I designed the analytical approach, coordinated with the Heartland Alliance monitoring and evaluation team on data quality checks (deduplication, logic checks for date and categorical variables). I produced all figures and tables, wrote the original manuscript draft, and coordinated revisions with co-authors through to publication in *BMJ Global Health*.

Paper 2: Viral Load Testing and Suppression Among FSWs

For Paper 2, I led the statistical analysis, which involved processing and cleaning routine programmatic data from over 34,000 individuals, implementing mixed-effects logistic regression models with random intercepts for 16 care facilities. I produced all figures and tables, wrote the original manuscript draft, and coordinated revisions with co-authors through to publication in *PLoS ONE*. I also contributed to the study design and methodology, in close collaboration with the PhD supervisor and senior co-authors.

Paper 3: Evaluation of the R Basics and Beyond Bootcamp

For Paper 3, I was a co-founder and lead developer of the GRAPH Courses initiative, which served as the training platform for the R Basics and Beyond bootcamp. This involved developing the platform's pedagogical approach, the website and learning management system, and the curriculum and content for several bootcamps run by the network.

For the R Basics and Beyond bootcamp, I led the curriculum development and the live workshop facilitation for a majority of the cohorts.

For the evaluation study itself, I designed the post-course survey instrument, led the data analysis, and wrote the original manuscript draft.

Articles

The three articles that comprise this thesis are arranged thematically by their position in the HIV cascade and the data science workforce that supports it — prevention (Paper 1), treatment (Paper 2), and the training of health professionals to analyse data of this kind (Paper 3) — rather than in order of publication.

The full text of each article appears after its cover page in the assembled thesis. (See `scripts/assemble_thesis_pdf.R` for the post-render step that splices the article PDFs into the rendered thesis at the marked positions.)

Paper 1 — PrEP Continuation Among Key Populations

Title: Oral PrEP continuation rates among key populations in Nigeria: a retrospective cohort study of a large-scale HIV prevention programme

Authors: Nwosu KD, Kalaiwo A, Ng'ambi W, Oyelaran O, Umoh P, Estill J, Keiser O

Journal: *BMJ Global Health* 2025; 10(12): e018739

DOI: 10.1136/bmjgh-2024-018739

Status: Published, December 2025

[Article PDF inserted here at final assembly — marker: ARTICLE-INSERT-1 — source: prep_paper.pdf]

Paper 2 — Viral Load Testing and Suppression Among Female Sex Workers

Title: Factors associated with viral load testing and viral suppression among HIV-positive female sex workers in Nigeria

Authors: Nwosu KD, Kalaiwo A, Ngambi WF, Estill J, Omo-Emmanuel UK, Emmanuel G, Keiser O

Journal: *PLoS ONE* 2024; 19(5): e0304487

DOI: 10.1371/journal.pone.0304487

Status: Published, May 2024

[Article PDF inserted here at final assembly — marker: ARTICLE-INSERT-2 — source: viral_suppression_paper.pdf]

Paper 3 — Evaluation of the R Basics and Beyond Bootcamp

Title: R Basics and Beyond: Outcomes of a Cohort-Based Online Programming Course for Health Professionals

Authors: Nwosu KD, [co-authors to be finalized]

Target journal: *Cogent Education*

Status: Manuscript in preparation (draft dated 2026-04-26)

[Article PDF inserted here at final assembly — marker: ARTICLE-INSERT-3 — source: tgc_paper_2026_04_26.pdf]

Conclusions and Perspectives

Summary of Key Findings

The PrEP analysis examined continuation among 43,788 key population clients enrolled across seven Nigerian states between 2020 and 2023. The 6-month continuation rate was 11.5%, substantially lower than comparable estimates of 28% from Cameroon and 25% from Zambia (53,54). Continuation varied markedly across KP groups, with transgender individuals showing the lowest rate (3.5%; aOR 0.37), likely reflecting higher levels of stigma, mistrust of healthcare providers, and competing health priorities in this group. Older clients had higher continuation than younger clients, and unemployment was associated with a small but significant reduction in continuation, suggesting that economic insecurity may negatively impact the sustaining of PrEP use.

The viral suppression analysis examined viral load testing and suppression among 34,976 HIV-positive female sex workers initiating ART in Nigeria between 2016 and 2022. Both indicators were high: 97.1% received at least one viral load test, and 94.5% were virally suppressed at the test closest to six months post-initiation, approaching the UNAIDS target of 95% suppression among people on ART (1). The odds of both outcomes increased for FSWs initiating ART in more recent years, consistent with improvements in the programme's reach and provision of care over the study period. Advanced WHO clinical stage at baseline was associated with lower odds of suppression, pointing to the importance of early diagnosis and ART initiation.

The R course evaluation assessed the R Basics and Beyond bootcamp across nine cohorts of 458 health professionals enrolled between 2023 and 2025. The completion rate of 55.9% substantially exceeded the single-digit rates typical of self-paced MOOCs (43), and 96% of post-course survey respondents gave a recommendation score of 9 or 10 out of 10. Pre-recorded video lessons (mean rating 9.3 out of 10) and live workshop facilitator interaction (9.2) were the highest-rated course components, highlighting the value of both asynchronous content delivery and synchronous cohort interaction. Time constraints (34%) and competing work priorities (30%) were the most frequently cited barriers, underscoring the limitation of a single fixed cohort cadence.

The HIV Cascade for Key Populations: Treatment Success, Prevention Gap

Juxtaposing the findings from the PrEP and viral suppression analyses reveals a striking asymmetry in the Nigerian KP programme's performance. On the treatment side, the programme has achieved remarkable success: once FSWs are diagnosed and linked to care, nearly all receive viral load testing and the vast majority achieve viral suppression. This demonstrates that community-based, KP-friendly service delivery models can overcome many of the structural barriers that typically impede care cascade progression for marginalized populations.

On the prevention side, the picture is far less encouraging. The very low PrEP continuation rates suggest that while the provider-facilitated approach is effective at promoting initial uptake, it has not succeeded in translating that uptake into sustained use. This disconnect echoes patterns seen in other PrEP programmes globally (35) and points to a fundamental challenge: the factors that drive individuals to start PrEP may be quite different from those required to maintain it over time.

Implications for HIV Treatment and Viral Suppression

The high rates of viral load testing and suppression observed in the viral suppression analysis are encouraging, but several findings warrant attention for continued programme improvement. First, the association between advanced WHO clinical stage at baseline and lower odds of viral suppression reinforces the well-established finding that late ART initiation compromises treatment outcomes (55). This finding is particularly relevant for FSWs, who face documented barriers to early HIV diagnosis and care entry, including fear of disclosure, stigma, and distrust of healthcare providers (56). Strategies to promote earlier diagnosis—such as community-based testing, HIV self-testing, and integration of testing with other services used by FSWs—may help address this gap.

Second, the association between education level and viral load testing adherence, though modest in magnitude, highlights the potential value of health literacy interventions for FSWs with limited formal education. Programmes might consider integrating simplified health education materials and adherence support into routine care, particularly for clients with lower educational attainment.

Third, the steady improvement in both testing and suppression rates over the study period suggests that the programme's quality improvement efforts have been effective. Identifying and documenting the specific programmatic changes that drove these improvements, such as expansion of viral load testing capacity, transition to more effective ART regimens (notably the shift from efavirenz- to dolutegravir-based regimens), and refinement of community-based service delivery, would be valuable for guiding similar programmes in other settings.

Addressing the PrEP Continuation Crisis

The very low PrEP continuation rates observed in the PrEP analysis demand urgent attention. Several strategies merit consideration.

First, the optimal balance between breadth of PrEP initiation and depth of pre-initiation engagement remains unresolved. The provider-facilitated approach used in the KP-CARE-1 programme prioritises uptake by offering PrEP to all eligible HIV-negative clients; an alternative would be to incorporate more thorough readiness assessment and motivational counselling at initiation, in the hope of identifying clients more likely to sustain use [verify and cite]. However, a recent cluster-randomised trial in western Kenya found that universal and risk-based PrEP delivery during pregnancy yielded comparable uptake and continuation (57), suggesting that broad initiation followed by client-led decisions on

continuation may be equally effective. More evidence from key population settings is needed to determine which approach best supports sustained use.

Second, the stark disparities across KP groups call for group-specific interventions. Transgender individuals, for example, had continuation rates nearly four times lower than FSWs. For transgender individuals, addressing the intersecting challenges of stigma, discrimination, competing health priorities (such as gender-affirming care), and healthcare avoidance may require dedicated service delivery strategies, including transgender-competent providers and peer support (58).

Third, the association between unemployment and lower PrEP continuation suggests that economic barriers, including transport costs, lost wages, and pharmacy fees, may disrupt the routines required for sustained PrEP use. Similar patterns have been documented elsewhere: in Namibia, unemployed clients had substantially lower odds of remaining in PrEP care (59), and a meta-ethnographic review of young men who have sex with men identified financial burdens as barriers to both adherence and retention (60). Programme adaptations such as multi-month dispensing, mobile or community-based refill options, and integration of PrEP services with economic support programmes could help address these barriers (61).

Fourth, it is important to recognize that low continuation does not necessarily equate to programme failure. Key populations often experience fluctuating HIV risk, and some clients may appropriately cycle on and off PrEP in response to changing risk behaviours (62,63). Future research should distinguish between discontinuation due to poor adherence and intentional, risk-informed cessation, as these have very different programmatic implications.

Finally, the emergence of long-acting injectable PrEP (cabotegravir, administered every two months (64)) and the dapivirine vaginal ring (65) as alternatives to daily oral PrEP may help address some of the barriers to continuation identified in this study. By reducing the burden of daily pill-taking, these newer modalities may improve continuation rates among key populations who struggle with the daily oral regimen.

Future Directions for HIV Research

For the HIV cascade studies, further research using more objective measures of HIV disease status (such as CD4 counts, which were unavailable in the current dataset) would strengthen the analysis of factors associated with viral suppression. Qualitative studies with both FSWs and healthcare providers could illuminate the specific mechanisms driving the observed improvements in testing and suppression rates over time, providing actionable insights for programme design. Additionally, extending the analysis to examine outcomes beyond 6-month viral suppression, including long-term retention in care, treatment switching, and mortality, would provide a more complete picture of the care cascade for this population.

For PrEP continuation research, qualitative studies that document clients' lived experiences with PrEP, their reasons for starting and stopping, and the contexts in which discontinuation occurs would provide rich interpretive context that quantitative analysis alone cannot capture. Studies that incorporate individual-level data on risk behaviours and biological adherence measures (such as drug-level monitoring) would similarly enrich interpretation of continuation patterns. The rollout of long-acting injectable PrEP in Nigeria and similar settings will also require dedicated evaluation to assess whether these newer modalities improve continuation among the KP groups identified as most vulnerable in this study.

Building Data Science Capacity for Global Health

Several aspects of the R course evaluation findings deserve further discussion.

The high ratings of both asynchronous (video lessons; mean 9.3 out of 10) and synchronous (live workshops; 9.2) components support the pedagogical theory underlying the flipped classroom model: that learning is optimised when self-paced content delivery is paired with guided practice and instructor interaction (47). The particularly high value placed on instructor support during live workshops suggests that for programming education, where learners frequently encounter frustrating errors that are difficult to resolve independently, real-time human guidance remains indispensable.

The finding that time constraints and competing work priorities were the primary barriers to completion highlights a fundamental tension in professional education: the working professionals who most need these skills are often those least able to commit sustained time to structured coursework. This tension has design implications for any data science training aimed at working health professionals. Programmes may benefit from modular formats that allow learners to drop in and out across weeks, built-in catch-up windows that reduce the cost of falling behind, and the offering of multiple tracks at different intensities so learners can match the workload to their capacity. Stronger asynchronous scaffolding—exercises that learners can complete and revisit independently—may also reduce the consequences of missing a synchronous session.

The Graph Courses: Broader Impact and Lessons Learned

As of early 2026, The Graph Courses has served over 9,000 learners from more than 60 countries across its self-paced and cohort-based offerings, with over 500 graduates from the intensive live bootcamps. Beyond these aggregate measures, the platform's development during this doctoral period offers two broader observations for scaling data science training in global health.

First, our experience suggests that a shared training infrastructure can host content from multiple partner organisations rather than each organisation building its own platform from scratch. The platform's portfolio has grown from a single R course to programmes spanning R, Python, generative AI, and statistics, and now hosts material from partner organisations including the World Health Organization (with courses on the Harmonized

Health Facility Assessment and a four-course Health Inequality Monitoring sequence) and the University of Oxford (a course on introductory statistics). In these cases, partners contribute domain content while the platform provides hosting, instructional design support, and delivery infrastructure. The number of partnerships is still small, so this is preliminary rather than definitive evidence; nonetheless, the early experience suggests that the approach may reduce duplication of effort and let partner organisations focus on their domain expertise rather than on platform development.

Second, the combination of free self-paced courses, low-cost live bootcamps, and freely available video content has helped balance accessibility with sustainability. The free self-paced tracks, which have reached thousands of learners globally, lower the barrier to entry and serve learners who cannot afford tuition or who prefer self-directed study. The platform's YouTube channel (youtube.com/@thegraphcourses) has accumulated over 163,000 views across 113 videos, providing additional open-access educational content—though view counts are an imperfect proxy for engagement. The live bootcamps, supported by a model that combines grant funding with course revenue, enable the platform to maintain low fees and provide substantial financial aid to learners from lower-resource settings. Together, these tracks allow the platform to serve the LMIC populations it aims to reach while generating the revenue needed to sustain operations and continued course development.

Future Directions for Data Science Training

For the training programme itself, longitudinal follow-up studies tracking whether course completion translates into sustained R use and improved work outputs in professional practice would strengthen the evidence base for this training model. Controlled study designs comparing cohort-based and purely self-paced approaches are a valuable future direction. Such a comparison was in fact the original research plan for this thesis component but was later adapted to an observational design as the programme scaled.

More broadly, The Graph Courses platform is positioned for continued expansion along several axes. Thematically, the addition of courses in advanced statistics, machine learning, and causal inference would extend the curriculum beyond introductory data skills into the more specialised competencies increasingly demanded by global health research. Institutionally, the hosting of additional external courses from partner organisations, building on the existing collaborations with WHO and the University of Oxford, offers the potential to establish the platform as a shared infrastructure for data science training across the global health sector.

Limitations

Several limitations should be considered when interpreting the findings of this thesis.

For the HIV studies (Papers 1 and 2), the observational design precludes causal inference. Both studies relied on routinely collected programmatic data, which may be subject to measurement error, incomplete recording, and other data quality issues that the study

team could not fully assess (66). The analyses may have been affected by unmeasured confounding; individual-level socioeconomic status, substance use, and psychosocial factors were not available in the dataset but likely influence both treatment and prevention outcomes. A further limitation is the analytical distance from the contexts in which the data were generated. The interpretation of routine programmatic data benefits from the qualitative context held by frontline staff and clients—the operational realities, local barriers, and informal practices that shape what is recorded and what is not. A short site visit during the doctoral period offered useful but limited insight into this context, and quantitative findings of this kind are best interpreted alongside in-depth knowledge of the programme's day-to-day operations. This points to the broader value of strengthening data science skills among in-country staff who already hold this contextual knowledge, complementing rather than substituting for analyses conducted from external academic settings. Finally, the findings from the Nigerian KP programme may not be directly generalisable to key populations in other settings with different epidemiological, legal, and service delivery contexts.

For the R course evaluation, the absence of a comparison group prevents conclusions about whether the cohort-based format itself drives higher completion rates compared to alternative approaches. Survey respondents over-represented course completers, which likely biases satisfaction estimates upward. The study relied on self-reported data subject to recall and social desirability bias. No longitudinal follow-up was conducted to assess long-term skill retention or professional application. As authors affiliated with the organisation that developed and delivers the course, we have aimed to report findings transparently, but our interpretation may still be subject to favourable bias.

Conclusion

The three studies in this thesis are united by the premise that data-driven decision-making can improve health outcomes for vulnerable populations, but only if both the data and the skills to analyse it are in place. Papers 1 and 2 demonstrate this premise in action: routine programmatic data, subjected to rigorous analysis, revealed a critical gap in PrEP continuation (the PrEP analysis), particularly among transgender individuals and younger clients, while also showing that Nigeria's KP programme is achieving strong treatment outcomes for FSWs on the care cascade (the viral suppression analysis). These findings can directly inform programme design, but they required analytical skills that remain scarce in the settings where such programmes operate. And because the rigorous interpretation of programmatic data depends on local knowledge of the system that produced it, building these skills among in-country staff,who already hold that knowledge,would strengthen not only the volume but also the quality of the resulting evidence.

The R course evaluation addresses this scarcity. By evaluating an online training programme that teaches health professionals the very skills used to produce the evidence in the PrEP and viral suppression analyses, it closes the loop: from identifying health challenges through data analysis, to building the workforce that can perform such analysis

at scale. The high completion rates and participant satisfaction suggest that cohort-based online training is a viable model for meeting this need.

Ultimately, this thesis argues that strengthening the HIV response for key populations, and improving global health more broadly, depends not only on better interventions and better data, but on expanding who has the ability to turn data into evidence.

Bibliography

1. UNAIDS. AIDS, crisis and the power to transform — 2025 global AIDS update [Internet]. Geneva, Switzerland: Joint United Nations Programme on HIV/AIDS; 2025. Available from: https://www.unaids.org/sites/default/files/2025-07/2025-global-aids-update-JC3153_en.pdf

2. Institute for Health Metrics and Evaluation. GBD Results Tool [Internet]. 2023. Available from: https://vizhub.healthdata.org/gbd-results/

3. Hay SI, Ong KL, Santomauro DF, et al. Burden of 375 diseases and injuries, risk-attributable burden of 88 risk factors, and healthy life expectancy in 204 countries and territories, including 660 subnational locations, 1990–2023: A systematic analysis for the Global Burden of Disease Study 2023. The Lancet. 2025;406(10513):1873–922.

4. Joint United Nations Programme on HIV/AIDS (UNAIDS). HIV estimates with uncertainty bounds, 1990–2025 [Internet]. Spreadsheet; 2025. Available from: https://www.unaids.org/en/resources/documents/2025/HIV_estimates_with_uncertainty_bounds_1990-present

5. Mboup S, Musonda R, Mhalu F, Essex M. HIV/AIDS. In: Jamison DT, Feachem RG, Makgoba MW, Bos ER, Baingana FK, Hofman KJ, et al., editors. Disease and mortality in sub-saharan africa [Internet]. 2nd ed. Washington, DC: The International Bank for Reconstruction; Development / The World Bank; 2006. Available from: https://www.ncbi.nlm.nih.gov/books/NBK2289/

6. World Bank. Southern Africa HIV/AIDS and tuberculosis plan of action, 2010–2011 [Internet]. Washington, DC: The World Bank; 2010. Available from: https://documents1.worldbank.org/curated/en/568641468209966098/pdf/643430WP0South00Box0362535B0PUBLIC0.pdf

7. Deeks SG, Lewin SR, Havlir DV. The end of AIDS: HIV infection as a chronic disease. The Lancet. 2013;382(9903):1525–33.

8. UNAIDS. Indicators and questions for monitoring progress on the 2021 political declaration on HIV and AIDS: Global AIDS Monitoring 2023. Geneva, Switzerland: Joint United Nations Programme on HIV/AIDS; 2022.

9. Korenromp EL, Sabin K, Stover J, Brown T, Johnson LF, Martin-Hughes R, et al. New HIV infections among key populations and their partners in 2010 and 2022, by world region: A multisources estimation. JAIDS Journal of Acquired Immune Deficiency Syndromes. 2024;95(1S):e34–45.

10. Baral S, Beyrer C, Muessig K, Poteat T, Wirtz AL, Decker MR, et al. Burden of HIV among female sex workers in low-income and middle-income countries: A systematic review and meta-analysis. The Lancet Infectious Diseases. 2012;12(7):538–49.

11. UNAIDS. The urgency of now: AIDS at a crossroads — 2024 global AIDS update [Internet]. Geneva, Switzerland: Joint United Nations Programme on HIV/AIDS; 2024. Available from: https://www.unaids.org/sites/default/files/media_asset/2024-unaids-global-aids-update_en.pdf

12. Schwartz SR, Nowak RG, Orazulike I, Keshinro B, Ake J, Kennedy S, et al. The immediate effect of the Same-Sex Marriage Prohibition Act on stigma, discrimination, and engagement on HIV prevention and treatment services in men who have sex with men in Nigeria: Analysis of prospective data from the TRUST cohort. The Lancet HIV. 2015;2(7):e299–306.

13. Shannon K, Strathdee SA, Goldenberg SM, Duff P, Mwangi P, Rusakova M, et al. Global epidemiology of HIV among female sex workers: Influence of structural determinants. The Lancet. 2015;385(9962):55–71.

14. Scorgie F, Chersich MF, Ntaganira I, Gerbase A, Lule F, Lo YR. Socio-demographic characteristics and behavioral risk factors of female sex workers in sub-Saharan Africa: A systematic review. AIDS and Behavior. 2012;16(4):920–33.

15. Kavanagh MM, Agbla SC, Joy M, Aneja K, Pillinger M, Case A, et al. Law, criminalisation and HIV in the world: Have countries that criminalise achieved more or less successful pandemic response? BMJ Global Health. 2021;6(8):e006315.

16. Vasireddy V, Brown NE, Shah N, Crowell TA. Sustaining HIV service delivery to key population clients using client-centred models during the debate and enactment of the Anti-Homosexuality Act in Uganda. Journal of the International AIDS Society. 2024;27(5):e26253.

17. World Health Organization. Consolidated guidelines on HIV, viral hepatitis and STI prevention, diagnosis, treatment and care for key populations [Internet]. Geneva, Switzerland: World Health Organization; 2022. Available from: https://www.who.int/publications/i/item/9789240052390

18. Ibiloye O, Masquillier C, Jwanle P, Van Belle S, Olmen J van, Lynen L, et al. Community-based ART service delivery for key populations in sub-Saharan Africa: Scoping review of outcomes along the continuum of HIV care. AIDS and Behavior. 2022;26(7):2314–37.

19. Kennedy CE, Yeh PT, Verster A, Luhmann N, Nguyen VTT, Mello MB de, et al. Do peer navigators improve initiation and retention in HIV/VH/STI treatment programs for people from key populations? A systematic review of effectiveness, values and preferences, and cost. JAIDS Journal of Acquired Immune Deficiency Syndromes. 2024;95(4):305–12.

20. UNAIDS. Nigeria [Internet]. 2024. Available from: https://www.unaids.org/en/regionscountries/countries/nigeria

21. Onovo AA, Adeyemi A, Onime D, Kalnoky M, Kagniniwa B, Dessie M, et al. Estimation of HIV prevalence and burden in Nigeria: A Bayesian predictive modelling study. eClinicalMedicine. 2023;62:102098.

22. Federal Ministry of Health, Nigeria. Nigeria HIV/AIDS Indicator and Impact Survey (NAIIS) 2018: Technical report. Abuja, Nigeria: Federal Ministry of Health; 2019.

23. Emmanuel F, Ejeckam CC, Green K, Adesina AA, Aliyu G, Ashefor G, et al. HIV epidemic among key populations in Nigeria: Results of the integrated biological and behavioural surveillance survey (IBBSS), 2020–2021. Sexually Transmitted Infections. 2024;101(1):10–6.

24. Human Dignity Trust. Nigeria [Internet]. 2026. Available from: https://www.humandignitytrust.org/country-profile/nigeria/

25. International Drug Policy Consortium, Youth RISE Nigeria. Harm reduction landscape and drug laws in Nigeria [Internet]. International Drug Policy Consortium; 2025. Available from: https://idpc.net/publications/2025/06/harm-reduction-landscape-and-drug-laws-in-nigeria

26. Platt L, Grenfell P, Meiksin R, Elmes J, Sherman SG, Sanders T, et al. Associations between sex work laws and sex workers' health: A systematic review and meta-analysis of quantitative and qualitative studies. PLOS Medicine. 2018;15(12):e1002680.

27. Ochonye B, Emmanuel G, Abang R, Sanni OF, Umoh P, Kalaiwo A, et al. Prevalence and factors associated with psychological distress among key populations in Nigeria. PLoS ONE. 2024;19(4):e0300901.

28. Nwosu KD, Kalaiwo A, Ng'ambi W, Oyelaran O, Umoh P, Estill J, et al. Oral PrEP continuation rates among key populations in Nigeria: A retrospective cohort study of a large-scale HIV prevention programme. BMJ Global Health. 2025;10(12):e018739.

29. Onovo A, Kalaiwo A, Agweye A, Odezulu E, Ene O, Odoh I, et al. Diagnosis and case finding according to key partner risk populations of people living with HIV in Nigeria: A retrospective analysis of community-led index partner testing services. eClinicalMedicine. 2022;43:101265.

30. Nwosu KD, Kalaiwo A, Ngambi WF, Estill J, Omo-Emmanuel UK, Emmanuel G, et al. Factors associated with viral load testing and viral suppression among HIV-positive female sex workers in Nigeria. PLoS ONE. 2024;19(5):e0304487.

31. Schaefer R, Schmidt HMA, Ravasi G, et al. Adoption of guidelines on and use of oral pre-exposure prophylaxis: A global summary and forecasting study. The Lancet HIV. 2021;8(8):e502–10.

32. Grant RM, Lama JR, Anderson PL, et al. Preexposure chemoprophylaxis for HIV prevention in men who have sex with men. New England Journal of Medicine. 2010;363(27):2587–99.

33. McCormack S, Dunn DT, Desai M, et al. Pre-exposure prophylaxis to prevent the acquisition of HIV-1 infection (PROUD): Effectiveness results from the pilot phase of a pragmatic open-label randomised trial. The Lancet. 2016;387(10013):53–60.

34. Molina JM, Charreau I, Spire B, et al. Efficacy, safety, and effect on sexual behaviour of on-demand pre-exposure prophylaxis for HIV in men who have sex with men: An observational cohort study. The Lancet HIV. 2017;4(9):e402–10.

35. Zhang J, Li C, Xu J, Hu Z, Rutstein SE, Tucker JD, et al. Discontinuation, suboptimal adherence, and reinitiation of oral HIV pre-exposure prophylaxis: A global systematic review and meta-analysis. The Lancet HIV. 2022;9(4):e254–68.

36. National Agency for the Control of AIDS (NACA). National HIV and AIDS strategic framework 2021–2025 [Internet]. Abuja, Nigeria: National Agency for the Control of AIDS; 2022. Available from: https://naca.gov.ng/wp-content/uploads/2022/03/National-HIV-and-AIDS-Strategic-Framework-2021-2025-Final.pdf

37. Gardner EM, McLees MP, Steiner JF, del Rio C, Burman WJ. The spectrum of engagement in HIV care and its relevance to test-and-treat strategies for prevention of HIV infection. Clinical Infectious Diseases. 2011;52(6):793–800.

38. Nnko S, Kuringe E, Nyato D, Drake M, Casalini C, Shao A, et al. Determinants of access to HIV testing and counselling services among female sex workers in sub-Saharan Africa: A systematic review. BMC Public Health. 2019;19(1):15.

39. Kiosia A, Boylan S, Retford M, Marques LP, Bueno FTC, Kirima C, et al. Current data science capacity building initiatives for health researchers in LMICs: Global & regional efforts. Frontiers in Public Health. 2024;12:1418382.

40. Forsetlund L, Bjørndal A, Rashidian A, Jamtvedt G, O'Brien MA, Wolf F, et al. Continuing education meetings and workshops: Effects on professional practice and health care outcomes. Cochrane Database of Systematic Reviews. 2009;(2):CD003030.

41. Bonkoungou B, Mosha F, Abianuru A, Okeibunor J, Utunen H, Balaciano G, et al. The evolutionary journey to a new normal for learning and capacity building of healthcare

workers to prepare and respond to health emergencies across Africa. Frontiers in Public Health. 2025;12:1455444.

42. Pappano L. The Year of the MOOC. The New York Times [Internet]. 2012 Nov; Available from: https://www.nytimes.com/2012/11/04/education/edlife/massive-open-online-courses-are-multiplying-at-a-rapid-pace.html

43. Jordan K. Initial trends in enrolment and completion of massive open online courses. The International Review of Research in Open and Distributed Learning. 2014;15(1):133–60.

44. Khalil H, Ebner M. MOOCs completion rates and possible methods to improve retention — a literature review. In: Proceedings of EdMedia 2014 — world conference on educational media and technology. Tampere, Finland: Association for the Advancement of Computing in Education (AACE); 2014. p. 1305–13.

45. Kizilcec RF, Pérez-Sanagustín M, Maldonado JJ. Self-regulated learning strategies predict learner behavior and goal attainment in Massive Open Online Courses. Computers & Education. 2017;104:18–33.

46. Reich J, Ruipérez-Valiente JA. The MOOC pivot. Science. 2019;363(6423):130–1.

47. Bishop JL, Verleger MA. The flipped classroom: A survey of the research. In: 2013 ASEE annual conference & exposition. Atlanta, Georgia; 2013. p. 23.1200.1–18.

48. Shapiro HB, Lee CH, Wyman Roth NE, Li K, Çetinkaya-Rundel M, Canelas DA. Understanding the massive open online course (MOOC) student experience: An examination of attitudes, motivations, and barriers. Computers & Education. 2017;110:35–50.

49. Dhawan S. Online learning: A panacea in the time of COVID-19 crisis. Journal of Educational Technology Systems. 2020;49(1):5–22.

50. Botero-Mesa S, Codeço Coelho F, Nwosu K, Wicht B, Venkatasubramanian A, Wagner O, et al. Leveraging human resources for outbreak analysis: Lessons from an international collaboration to support the sub-Saharan African COVID-19 response. BMC Public Health. 2022;22(1):1073.

51. Nwosu KD, Vaz J, McKinley A, Valle Campos A, Merzouki A, Nguimbis B, et al. Introduction to data analysis with R [Internet]. The GRAPH Courses; 2025. Available from: https://datawithr.com

52. Nwosu KD, McKinley A, Valle Campos A, Merzouki A, Nguimbis B, Hsin B, et al. Introduction to data science with Python [Internet]. The GRAPH Courses; 2025. Available from: https://pythondatabook.com

53. Ndenkeh JJrN, Bowring AL, Njindam IM, et al. HIV pre-exposure prophylaxis uptake and continuation among key populations in Cameroon: Lessons learned from the CHAMP program. Journal of Acquired Immune Deficiency Syndromes. 2022;91(1):39–46.

54. Claassen CW, Mumba D, Njelesani M, et al. Initial implementation of PrEP in Zambia: Health policy development and service delivery scale-up. BMJ Open. 2021;11(7):e047017.

55. Kitahata MM, Gange SJ, Abraham AG, Merriman B, Saag MS, Justice AC, et al. Effect of early versus deferred antiretroviral therapy for HIV on survival. New England Journal of Medicine. 2009;360(18):1815–26.

56. Lancaster KE, Cernigliaro D, Zulliger R, Fleming PF. HIV care and treatment experiences among female sex workers living with HIV in sub-Saharan Africa: A systematic review. African Journal of AIDS Research. 2016;15(4):377–86.

57. Kinuthia J, Dettinger JC, Stern J, Ngumbau N, Ochieng B, Gomez L, et al. Risk-based versus universal PrEP delivery during pregnancy: A cluster randomized trial in Western Kenya from 2018 to 2019. Journal of the International AIDS Society. 2023;26(2):e26061.

58. Teng LA, Sha Y, Fletcher LM, et al. Barriers to uptake of PrEP across the continuum among transgender women: A global scoping review. International Journal of STD & AIDS. 2023;34(5):299–314.

59. Kosmas K, Moyo E, Hliongwa M, et al. Retention of clients in HIV oral pre-exposure prophylaxis care in Engela, Namibia. African Journal of Primary Health Care & Family Medicine. 2025;17(1):e1–8.

60. Tuan LA, Pham LQ, Khuyen TT, et al. Facilitators and barriers to HIV pre-exposure prophylaxis adherence and retention among young men who have sex with men: A meta-ethnographic scoping review. AIDS and Behavior. 2025;29(4):1075–88.

61. Haberer JE, Mujugira A, Mayer KH. The future of HIV pre-exposure prophylaxis adherence: Reducing barriers and increasing opportunities. The Lancet HIV. 2023;10(6):e404–11.

62. Reed JB, Shrestha P, Were D, et al. HIV PrEP is more than ART-lite: Longitudinal study of real-world HIV PrEP services identifies missing measures meaningful to HIV prevention programming. Journal of the International AIDS Society. 2021;24(10):e25827.

63. Fazito E, Cuchi P, Mahy M, et al. Analysis of duration of risk behaviour for key populations: A literature review. Sexually Transmitted Infections. 2012;88(Suppl 2):i24–32.

64. Landovitz RJ, Donnell D, Clement ME, Hanscom B, Cottle L, Coelho L, et al. Cabotegravir for HIV prevention in cisgender men and transgender women. New England Journal of Medicine. 2021;385(7):595–608.

65. Baeten JM, Palanee-Phillips T, Mgodi NM, Mayo AJ, Szydlo DW, Ramjee G, et al. Safety, uptake, and use of a dapivirine vaginal ring for HIV-1 prevention in African women (HOPE): An open-label, extension study. The Lancet HIV. 2021;8(2):e87–95.

66. Gloyd S, Wagenaar BH, Woelk GB, Kalibala S. Opportunities and challenges in conducting secondary analysis of HIV programmes using data from routine health information systems and personal health information. Journal of the International AIDS Society. 2016;19(5(Suppl 4)):20847.